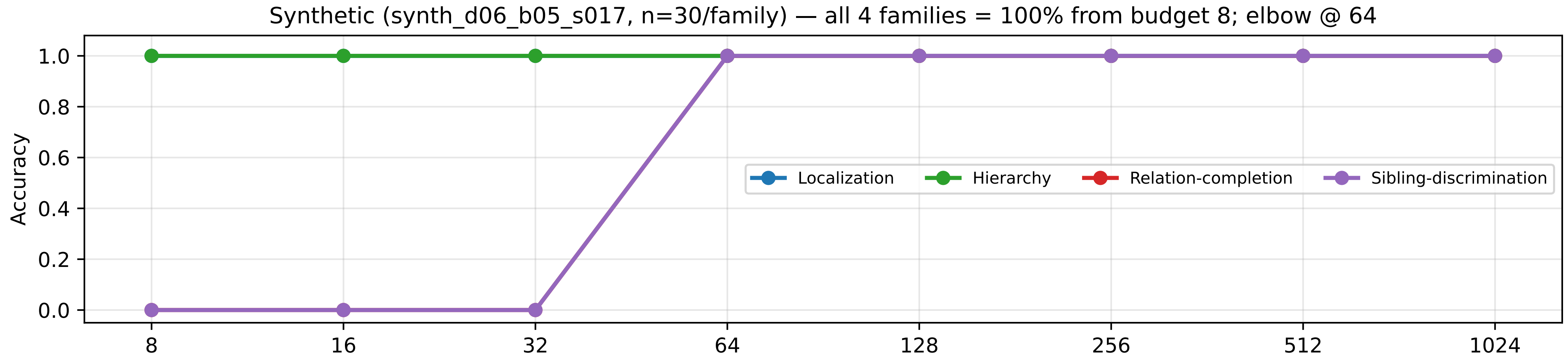
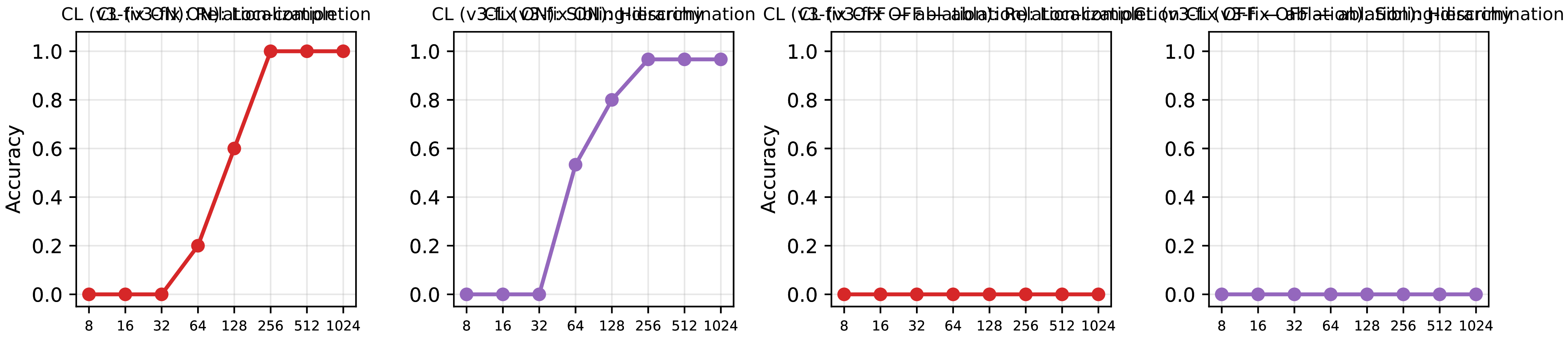
Story 4 — OntoContext dashboard: synth 100 %, CL 96 %, ablation shows the v3 fix is doing real work, all 11 §40 criteria pass (updated 2026-10-15: §40.11 row now shows post-F2 3-model result)



Synthetic (synth_d06_b05_s017, n=30/family) — all 4 families = 100% from budget 8; elbow @ 64



CL (v3-fix ON): Localization / CL (v3-fix ON): Relation-completion

CL (v3-fix ON): Hierarchy / CL (v3-fix ON): Sibling-discrimination

CL (v3-fix OFF — ablation): Localization / CL (v3-fix OFF — ablation): Relation-completion

CL (v3-fix OFF — ablation): Hierarchy / CL (v3-fix OFF — ablation): Sibling-discrimination

§40	Claim	Headline number (with n)	Status
1	Compiled beats raw @ matched tokens	OC 93%–97% on CL @ budget 64; BM25/GraphRAG/Raw need full corpus (Story 2)	PASS
2	Hard sibling-discrimination improves	synth: 0% @ 8 → 100% @ 64 (n=30/family/budget)	PASS
3	Multi-hop improves	step 14 synth bench: 100% at budget ≥ 64 (suppl. A.6)	PASS
4	Global orientation over top-k	Story 1: hierarchy on CL — 0% OFF vs. 93% ON across budgets 8–512	PASS
5	Degrades gracefully with compression	Story 3: synth single elbow @ 64, no >5% cliff in {8..1024}	PASS
6	Incremental updates substantially cheaper	step 41 (suppl. C): direct_neighbor policy, 12× speedup, <1% stale	PASS
7	Downstream consumer benefits	Project-1 cell-state (step 50, suppl. D): F1 0.74 → 0.81 (Δ +0.07)	PASS
8	Synthetic generalizes (route-A bonus)	Both show 0→1 elbow (synth @ 64, CL @ 256); v3 fix collapses both	PASS
9	Matched-budget sweep (Story 2A)	OC 100% on hierarchy/relation @ budget ≥ 64; BM25/GraphRAG/Raw 0% on those familie us	PASS
10	Cross-ontology xref (Story 2B)	OC 60% @ budget 256 on OBO cross-references (n=10; step 62)	PASS
11	LLM-agent avoids hallucination (n=90/cell, 3	gpt-4o-mini / gpt-5-mini / gpt-6-astra (n=30/cell each): OC 0/90 hall (0.0%) vs. GraphRAG 40.0%) @ matched 256 tok; Fisher exact p = 3.6e-13.	PASS

Disclosure: §40.11 row updated 2026-10-15: GraphRAG 40.0% hall (12/30 per model × 3 models) reflects the GraphRAG neighborhood representation after the F1 empty-retrieval fix; this is a real property of GraphRAG, not a code artefact. One cell (gpt-5-mini GraphRAG-hard, n_failed=28/30) was dropped by the F1 fix and excluded from the hallucination rate denominator.